

Mixed Hodge Structures First Talk Concise

1. Review

Let X be a smooth projective complex variety, or more generally a compact Kähler manifold. The classical theorem says that

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X), \quad \overline{H^{p,q}(X)} = H^{q,p}(X).$$

The point of this review is to recall where this decomposition comes from. There are three complementary descriptions:

- analytic: harmonic representatives;
- complex-geometric: closed forms with (p, q) representatives;
- algebraic: the Hodge filtration obtained from the filtered holomorphic de Rham complex

1.1 Dolbeault theorem

X be any complex manifold. The complex structure decomposes smooth forms into types: (as $\mathcal{C}^\infty$ -modules)

$$\mathcal{A}_X^k = \bigoplus_{p+q=k} \mathcal{A}_X^{p,q}, \quad d = \partial + \bar{\partial}.$$

The operator $\bar{\partial}$ has bidegree $(0, 1)$. Let Ω_X^p denote the sheaf of holomorphic p -forms (not a $\mathcal{C}^\infty$ -module). The Dolbeault resolution is

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,1} \xrightarrow{\bar{\partial}} \dots$$

Taking global sections gives

$$H^q(X, \Omega_X^p) \cong H^q(\Gamma(X, \mathcal{A}_X^{p,\bullet}), \bar{\partial}).$$

This is the Dolbeault theorem: sheaf cohomology of holomorphic forms is computed by smooth (p, q) -forms. It gives a useful interpretation of the Hodge pieces:

$H^{p,q}$ is represented by $\bar{\partial}$ -closed forms of type (p, q) , modulo $\bar{\partial}$ -exact forms.

1.2 Hodge theorem and Hodge decomposition

Let X be a compact Kähler manifold. On one hand, the smooth de Rham complex computes the same cohomology:

$$H_{\text{dR}}^n(X, \mathbb{C}) \cong H^n(X, \mathbb{C}).$$

On the other, choose a Kähler metric. The Hodge theorem (works for compact oriented Riemann mflds) identifies cohomology with harmonic forms:

$$H_{\text{dR}}^n(X, \mathbf{C}) \cong \mathcal{H}_d^n(X), \quad \mathcal{H}_d^n(X) := \ker(\Delta_d).$$

Every class therefore has a unique harmonic representative. The Kähler identities imply

$$\Delta_d = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}.$$

Because these Laplacians respect (p, q) -type,

$$\mathcal{H}_d^n(X) = \bigoplus_{p+q=n} \mathcal{H}^{p,q}(X).$$

Taking cohomology classes gives

$$H^n(X, \mathbf{C}) = \bigoplus_{p+q=n} H^{p,q}(X), \quad H^{p,q}(X) \cong H^q(X, \Omega_X^p).$$

Equivalently, every cohomology class can be represented by a closed form which is a sum of closed forms of pure types, and the harmonic representative makes this choice canonical. Complex conjugation exchanges (p, q) and (q, p) .

1.2.1 Hodge decomposition of an elliptic curve

Let

$$E = \mathbf{C}/\Lambda, \quad \Lambda = \mathbf{Z} + \mathbf{Z}\omega, \quad \text{Im}(\omega_2) > 0.$$

The forms dz and $d\bar{z}$ on $\mathbf{C}$ are invariant under translations by Λ , so they descend to global forms on E . They are translation-invariant, hence harmonic for the flat Kähler metric. Since E is a real two-dimensional torus,

$$H^1(E, \mathbf{C}) = \mathbf{C}[dz] \oplus \mathbf{C}[d\bar{z}].$$

The two summands have types $(1, 0)$ and $(0, 1)$:

$$H^{1,0}(E) = \mathbf{C}[dz] \cong H^0(E, \Omega_E^1), \quad H^{0,1}(E) = \mathbf{C}[d\bar{z}] \cong H^1(E, \mathcal{O}_E).$$

Thus $h^{1,0} = h^{0,1} = 1$. The top-degree form $dz \wedge d\bar{z}$ is a nonzero $(1, 1)$ -form, so

$$H^0(E, \mathbf{C}) = H^{0,0}(E), \quad H^2(E, \mathbf{C}) = H^{1,1}(E) = \mathbf{C}[dz \wedge d\bar{z}].$$

The Hodge diamond is therefore

$$\begin{array}{ccc} & 1 & \\ 1 & & 1 \\ & 1 & \end{array}$$

Remark. Every elliptic curve has the same Hodge numbers, namely $h^{1,0} = h^{0,1} = 1$, but not the same Hodge structure. In the presentation $E = \mathbf{C}/(\mathbf{Z} + \tau\mathbf{Z})$, the Hodge filtration on $H^1(E, \mathbf{C})$ remembers the period parameter τ : $F^1 H^1(E, \mathbf{C})$ is the line spanned by $[dz]$ inside the complexification of the lattice dual. Thus the Hodge structure varies with the complex structure of E ; at the rational level, isogenous elliptic curves give the same Hodge structure up to the corresponding rational isomorphism.

1.3 Hard Lefschetz and Lefschetz decomposition

Let X , compact Kähler, have dimension d , and let ω be a Kähler class. Cup product with ω defines

$$L : H^n(X, \mathbf{C}) \longrightarrow H^{n+2}(X, \mathbf{C}), \quad \alpha \longmapsto \omega \cup \alpha.$$

The hard Lefschetz theorem says that

$$L^{d-n} : H^n(X, \mathbf{C}) \xrightarrow{\sim} H^{2d-n}(X, \mathbf{C}) \quad (n \leq d).$$

The primitive cohomology is

$$H_{\text{prim}}^n(X) := \ker (L^{d-n+1} : H^n(X, \mathbf{C}) \rightarrow H^{2d-n+2}(X, \mathbf{C})).$$

Lefschetz decomposition expresses every class uniquely as

$$H^n(X, \mathbf{C}) = \bigoplus_{r \geq 0} L^r H_{\text{prim}}^{n-2r}(X).$$

1.4 Grothendieck's algebraic de Rham theorem

For a smooth algebraic variety X over $\mathbf{C}$, algebraic differential forms form the algebraic de Rham complex

$$\Omega_{X/\mathbf{C}}^\bullet : \mathcal{O}_X \xrightarrow{d} \Omega_{X/\mathbf{C}}^1 \xrightarrow{d} \Omega_{X/\mathbf{C}}^2 \xrightarrow{d} \cdots.$$

Grothendieck's algebraic de Rham theorem identifies its algebraic hypercohomology with the cohomology of the associated analytic space:

$$\mathbb{H}^n(X, \Omega_{X/\mathbf{C}}^\bullet) \cong H^n(X^{\text{an}}, \mathbf{C}).$$

For smooth projective X , filter this complex by form degree:

$$F^p \Omega_X^\bullet = [0 \rightarrow \cdots \rightarrow 0 \rightarrow \Omega_X^p \rightarrow \Omega_X^{p+1} \rightarrow \cdots].$$

The resulting spectral sequence is

$$E_1^{p,q} = H^q(X, \Omega_X^p) \implies H^{p+q}(X, \mathbf{C}).$$

Its degeneration at E_1 is the algebraic expression of the Hodge decomposition. The filtration is canonical and algebraic; the splitting into (p, q) -pieces uses the compact Kähler geometry.

2. Why usual Hodge decomposition fails for general varieties

A key consequence of Hodge decomposition is that odd cohomologies should have even dimension.

2.1 The open example: $\mathbf{G}_m$

Let $U = \mathbf{G}_m = \mathbf{P}^1 \setminus \{0, \infty\}$. Since $U \simeq S^1$ topologically,

$$H^1(U, \mathbf{Q}) \cong \mathbf{Q}.$$

The class is represented on $\mathbf{P}^1$ by the logarithmic form dz/z . We shall see later that it is of type $(1, 1)$, not of type $(1, 0)$ or $(0, 1)$.

2.2 The singular example: a nodal rational curve

Let C be an irreducible projective rational curve with one node. Its normalization

$$\nu : \mathbf{P}^1 \longrightarrow C$$

identifies two points. Topologically, this identifies two points of S^2 and creates one circle, so

$$H^1(C, \mathbf{Q}) \cong \mathbf{Q}.$$

We shall see later that its natural weight is 0.